

Computer Graphics

MID Lab Exam

(20 Marks) Time: 1.5 hours

1. A student is planning to draw a line using a computer graphics algorithm and found a initial pixel as (120,350) and final pixel as (530,780). In-order to implement the program with highest accuracy the student needs an efficient algorithm for the implementation. Using the code, the student also wants to print the value of y when the value of x is 500 and the slope of the line.
Construct the implementation of the algorithm with specific requirements keeping in mind about the efficiency of the implementation.

10 Marks

Sample: Enter the co-ordinates of initial pixel: 120 350

Enter the co-ordinates of the final pixel: 530 780

The value of y when x=500 is:

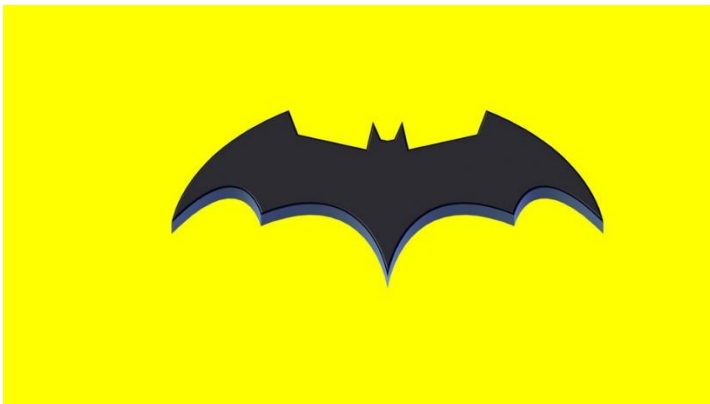
The slope of the line (m) is:

2. Create a image using OpenGL library from the following set A, B and C. Highlight the primitives used in the program using comments. Follow the set assigned to each student attached at the end of the drawings.

10 Marks

There are 3 sets of drawings assigned to each student:

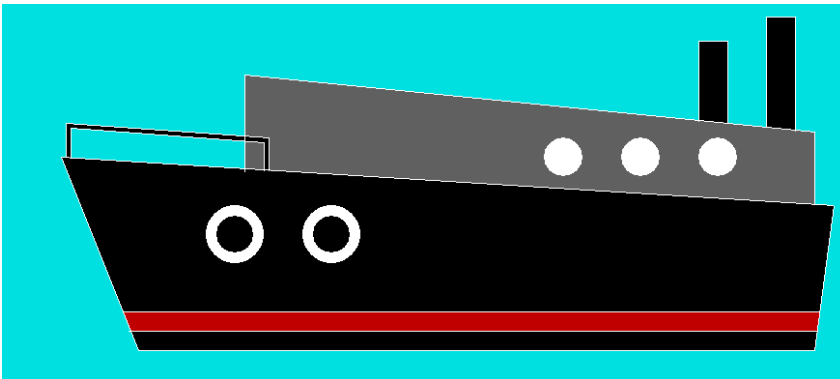
Set A: Create the following logo as close as possible.



Set B: Draw the following diagram as accurately as possible.



Set C: Draw the ship as accurately as possible.



Problems assigned as below:

ID	Name	SET
15-30410-3	RAHMAN, MD.ARIFUR	A
17-34443-1	RATRI, MST. SURAIYA	B
20-41868-1	SAKIB, MEHEDI HASAN	C
20-41906-1	ISLAM, MD. ARAF	A
20-42669-1	TANISHA, JANNATUL FERDOSE	B
20-43012-1	MAHMUD, IMTIAZ	C

20-43056-1	SAKIB, AN NAZMUS	A
20-44207-3	FAHIM, MIDDAT GHANI	B
21-44626-1	TAMAL, TANVIR HASAN	C
21-44953-2	CHOWDHURY, AHADUL ISLAM	A
21-44985-2	AHAD, NABILA ANJUM BENTA	B
21-45276-2	MASUD, SYED RAYHAN	C
21-45413-3	FIAZ-AL-FARDIN	A
21-45415-3	DIHAN, TANJIM ISRAK	B
21-45488-3	HASAN, A.S.M MAHMUDUL	A
21-45601-3	HADI, HANZALA HAMID	C
21-45676-3	RAFI, MD. HASIBUR RAHMAN	A
21-45830-3	ALAM, RAYAN	B
21-45981-3	BISWAS, SUDIPTA	C
22-46024-1	ORPITA, MAHNAZ TABASSUM	A
22-46066-1	HASAN, MD ZAFRUL	B
22-46256-1	HASAN, MD. MAHMUDUL	C
22-46259-1	HASAN, MAHEDI	A
22-46374-1	CHOWDHURY, MD. RAKIB	B
22-46424-1	ROY, PRITHOY CHANDRA	C
22-46459-1	NILOY, NAFIUR RAHMAN	A
22-46543-1	HASAN, NAHID	B
22-46591-1	SHIL, EMON KUMAR	C
22-46599-1	DAS, EMON	A
22-46605-1	HOSSAIN, MD. HASIBUL	A
22-46629-1	ARPON, SAJIN MAHMUD	A
22-46684-1	SUJON, MD. ATIK ISHRAK	A
22-46736-1	CHOWDHURY, UPANTA	B
22-46824-1	OLIVE, MAHRAJ AFSAN	B
22-46894-1	RAIYAN, ASHFI MUHAMMED	B
22-46929-1	SARKER, SOUMIK	C
22-46956-1	MUNSHI, MD. NAVID	C
22-46989-1	TRIPTY, FATEMA AKTER	C
22-46994-1	MAMUN, MD ABDULLAH AL	B
22-47045-1	SULTANA, TARIN	C